

Vansh Kumar

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EDUCATION

Delhi Technological University
B.Tech in Civil Engineering

2023 – 2027
CGPA: 7.0

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, HTML/CSS, JavaScript

Tools/Frameworks: Git, Linux (Ubuntu), Jupyter Notebook, Google Colab, Streamlit, PyTorch, React, FastAPI, Render, Vercel

Libraries: NumPy, pandas, Matplotlib, Scikit-learn

Core: Machine Learning, Data Analysis, Natural Language Processing, Problem Solving, Reinforcement Learning, Deep Learning, Full-Stack Development

PROJECTS

CyberArena RL — Multi-Agent Adversarial Reinforcement Learning | Python, PyTorch, React, FastAPI, NumPy
GitHub | Live | API

- Built adversarial cybersecurity simulation with attacker and defender agents co-evolving strategies using Deep Q-Network (DQN)
- Implemented PyTorch neural network ($29 \rightarrow 128 \rightarrow 128 \rightarrow 12$) with experience replay buffer (10k capacity) and target network stabilization
- Designed 12 attack types (Phishing, Ransomware, Zero-Day, DDoS etc.) and 12 defense mechanisms (IDS, Honeypot, Firewall etc.)
- Built pixel-art React frontend with live neural network visualizer showing neuron activations in real time
- Deployed full-stack: FastAPI backend on Render, React on Vercel, analytics dashboard on Streamlit Cloud

Spam Classifier | Python, NLP, Machine Learning, Streamlit
GitHub | Live | API

- Developed an NLP-based spam classification model achieving **90%+ accuracy** on SMS dataset (5,000+ messages)
- Built and deployed a Streamlit web app and REST API enabling real-time predictions with low latency
- Implemented end-to-end pipeline including text preprocessing, TF-IDF feature extraction, and model evaluation

Project Nova | Python, Data Analysis, Machine Learning
GitHub

- Designed a synthetic data generation system handling **10,000+ records** with a custom scoring model (Nova Score)
- Performed feature engineering and EDA to identify key patterns and correlations in data
- Built modular pipeline reducing code redundancy and improving scalability for future extensions

EXPERIENCE

AI/ML Contributor
NeuralAI Society, Delhi Technological University

Sept 2023 – April 2025

- Worked on machine learning projects and participated in technical workshops
- Collaborated on real-world AI applications and strengthened ML fundamentals

Sports Council, Delhi Technological University
Co-Head, Public Relations

Aug 2024 – Present

- Led PR strategies and coordinated cross-functional teams for large-scale university events to enhance campus-wide participation and visibility
- Managed end-to-end event promotions, including content creation, targeted outreach, and operational coordination

Public Relations Associate

Sept 2023 – April 2024

- Coordinated with 10+ teams to execute PR campaigns and support operations during various university events
- Designed promotional content that effectively contributed to increasing student engagement and event participation

CERTIFICATIONS

Python (Scaler) | C++ (Scaler) | Machine Learning (Scaler) | Cyber Security (Credly) | Revit MEP (SourceCAD)

ACHIEVEMENTS

- Ranked among **Top 100 teams out of 4000+ participants** in GrabHack Hackathon after qualifying multiple rounds